KARDIOGEN™

Your Heart disease genetic risk assessment report

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SAMPLE DETAILS

Full Name/Ref. No:	**Mr. Aromal P Rajagopal**	Order ID / Sample ID:	**511168/7713169**
Gender:	Male	Sample Type:	EDTA blood
Date of Birth/Age:	44 years	Date of Sample Collection:	NA
Referring Clinician:	Dr. Joshy EV, Genchrome - Bangalore	Date of Sample Receipt:	07th October 2022
Test Requested:	Coronary Artery Disease-Polygenic Risk Score Test (CAD-PRS)	Date of Order Booking:	08th October 2022
Test Code:	MGM1473	Date of Report:	21st October 2022

CLINICAL DIAGNOSIS/ SYMPTOMS/ HISTORY

Mr. Aromal P Rajagopal has a lipid profile of total cholesterol of 210 mg/dl, triglyceride level of 160 mg/dl, HDL level of 42 mg/dl, and LDL level of 136 mg/dl. He has a family history of diabetes mellitus in his father, mother and heart disease in father. He has been tested and evaluated for Coronary Artery Disease Polygenic Risk Score.

KNOW YOUR KARDIOGEN RISK SCORE

This report contains your KARDIOGEN Risk Score (KRS).

KARDIOGEN Risk Score is a genome-wide polygenic risk score which outlines your probability of developing Coronary artery disease (CAD) due to genetic factors. KARDIOGEN Risk Score is determined through an assessment of more than 6 million genetic variants. Based on your KARDIOGEN Risk Score, you are classified as to be at High, Moderate or Average Risk for developing CAD. This insight will help you take preventive actions for the True Wellbeing of your heart.

READING THE REPORT

To make the report simple and easy to use, certain terms that have been used are linked to their related explanations.

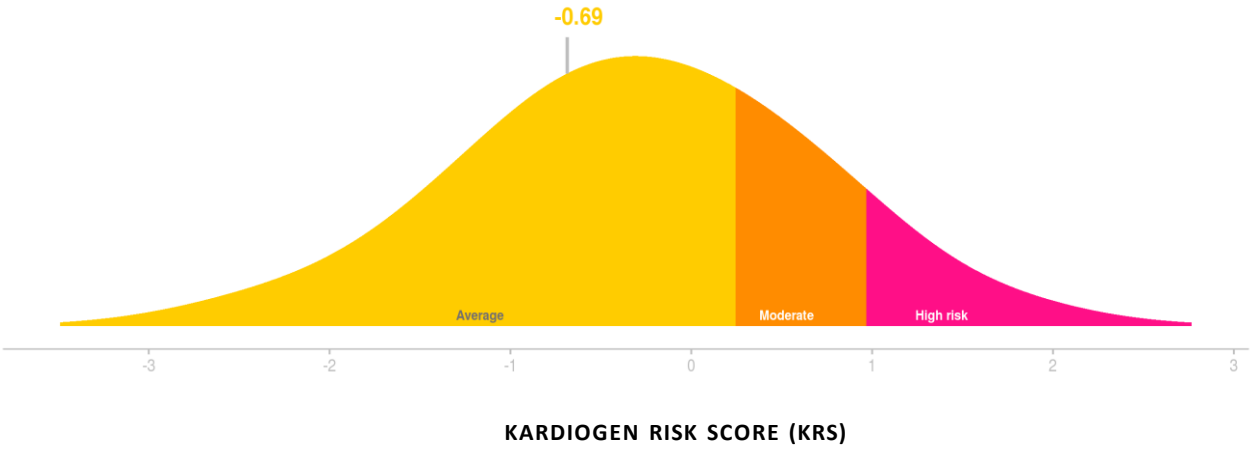
All words that are underlined in blue are clickable.

If you are reading the printed version of this report, go to the page number listed next to the term to view an explanation of the term.

Report summary - page 3
Heart disease - page 4
Average risk in South Asian population - page 6
Combined risk factors for heart disease - page 7
Test methodology - page 8

YOUR REPORT SUMMARY

KARDIOGEN RISK SCORE	-0.69
ODDS RATIO	<=1
RISK CATEGORY	Average



What to do next

Knowing that you are in the average-risk category is the first step towards managing your risk.

Your risk of developing heart disease is also influenced by other physiological/ lifestyle parameters such as blood pressure, cholesterol levels and how much you smoke. These physiological parameters constitute your clinical risk factors. A combination of clinical risk factors and your KARDIOGEN Risk Score determines your risk for developing heart disease.

While KARDIOGEN Risk Score is based on genetic factors and cannot be changed, the clinical risk factors can be managed through modifying lifestyle and taking medication.

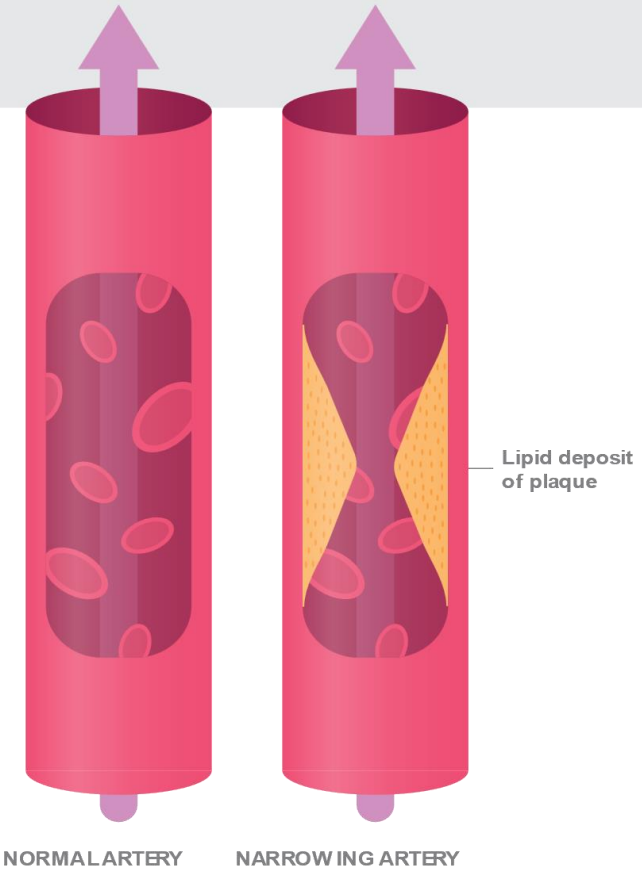
APPENDIX

In this section, we explain the terms underlined in blue in page 2

HEART DISEASE

When we talk about heart disease, we are specifically referring to Coronary Artery Disease (CAD), which is the most common type of heart disease.

Coronary Artery Disease (CAD) is a heart condition that affects the blood vessels which supply oxygenated blood to the heart. Plaque gets deposited on the walls of the arteries, which reduces the supply of oxygenated blood to the heart. The plaque blocks the arteries either partially or completely. This blockage can result in a heart attack.



KARDIOGEN RISK SCORE

There are millions of genetic variations, with individual minor effect, that affect your probability of developing heart disease.

We use the latest techniques to extract your DNA using the sample provided and analyse over 6 million markers in your DNA to arrive at your KARDIOGEN Risk Score.

KARDIOGEN Risk Score is therefore a genome-wide Polygenic Risk Score. This risk score is specific for CAD.

RISK CATEGORY BASED ON KARDIOGEN RISK SCORE

High risk	KRS>=0.973
Moderate risk	KRS 0.236-0.972
Average risk	KRS<=0.235

ODDS RATIO

We also provide a number that indicates your probability of developing heart disease based on your KARDIOGEN Risk Score. This number is your odds ratio. Your odds ratio is a multiplier indicating your chance of developing heart disease in comparison to the average population. For example, if your odds ratio is 2, then you are twice as likely to develop heart disease compared to the average population. If the number is 0.5 then you are half as likely to develop heart disease compared to the average population.

ODDS RATIO BASED ON RISK CATEGORY

High risk	OR>=2.43
Moderate risk	OR 1.19-1.67
Average risk	OR<=1

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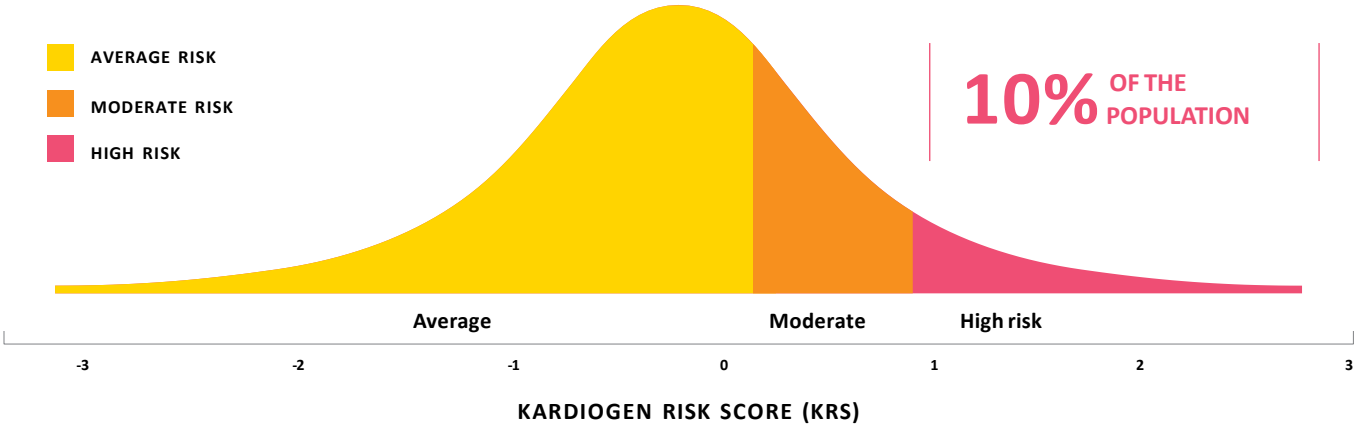
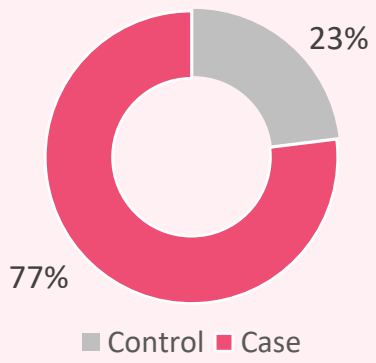
AVERAGE RISK IN SOUTH ASIAN POPULATION

THE GRAPH HERE EXPLAINS THE OVERALL DISTRIBUTION OF POPULATION BASED ON THEIR RISK SCORE.



KARDIOGEN Risk Score (KRS) distribution in the South Asian population from MedGenome's study cohort: Distribution of normalized KARDIOGEN Risk Score is given in the X-axis, based on MedGenome's study cohort, consisting of 1209 samples without heart disease belonging to South Asian ethnicity. Samples in the 10th decile (see Test methodology) with highest KARDIOGEN Risk Score, are highlighted in red colored area in the distribution. The KARDIOGEN Risk Score values in the red zone are >= 0.973 where 3.14 was the highest KARDIOGEN Risk Score seen in MedGenome's study cohort. Individuals in the 10th decile have an odds ratio of 2.43 (2.43 times more likely to develop CAD compared to healthy population). Individuals in the orange zone have an odds ratio ranging between 1.19 through 1.67 and have a moderate risk of developing CAD. Individuals in the yellow zone have a KARDIOGEN Risk Score <=0.235, with a population average polygenic risk for CAD, depending on their physiological status and family history of CAD.

Proportion of cases and control in the High-Risk group from our study population



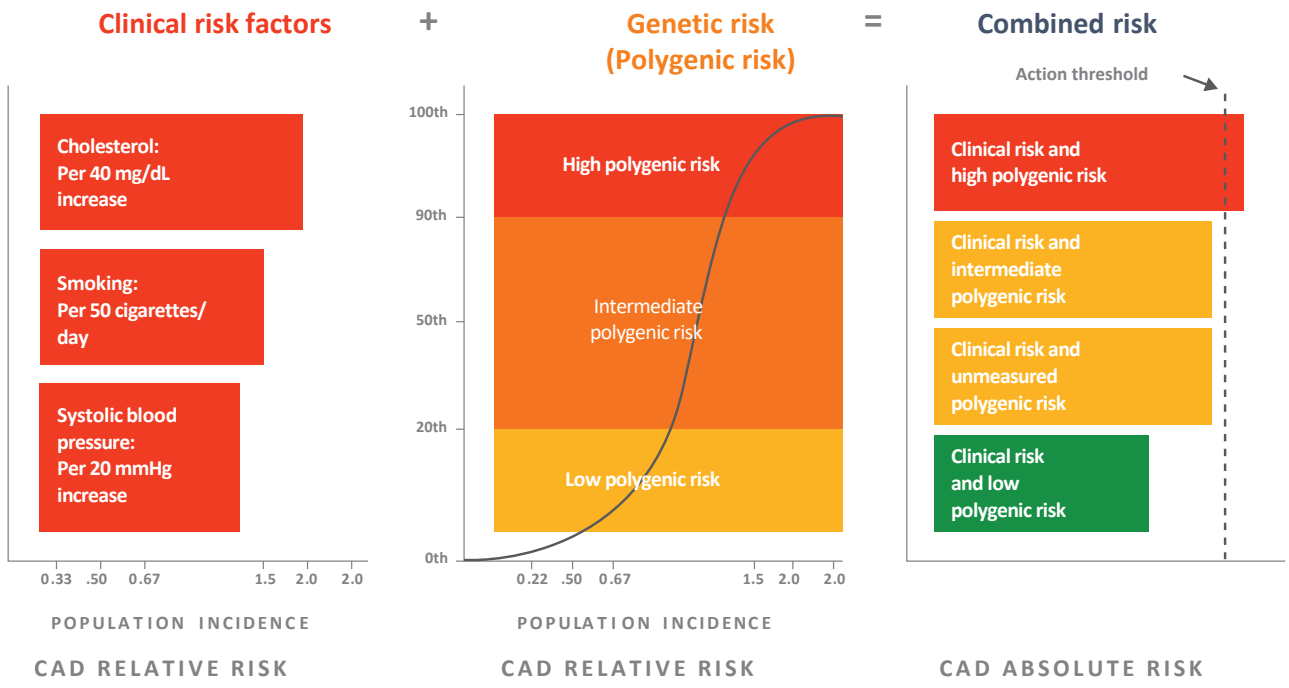
In the studied population, we have observed that 77% of the individuals in the High-Risk group (10th decile) of KARDIOGEN Risk Score, were clinically diagnosed to have CAD (cases). However, there were 23% of individuals without CAD (controls), at the time of participant recruitment for the study.

COMBINED RISK FACTORS FOR CAD

While your KARDIOGEN Risk Score reflects your hidden CAD status, your clinical risk factors - like your cholesterol level, blood pressure, blood sugar, triglycerides, the amount you smoke, and several other factors also influence the probability of developing heart disease.

Together, your KARDIOGEN Risk (Polygenic Risk) and your Clinical Risk factors form your Combined Risk that you need to be aware of and take actions accordingly.

Let us take a quick look at the diagram below to understand this better [2].



TEST METHODOLOGY

ARRAY BASED WHOLE GENOME GENOTYPING

Single nucleotide polymorphism genotyping was performed for the sample using SNP arrays. Specified quality and quantity of genomic DNA was amplified, enzymatically fragmented and precipitated using propanol, followed by re-suspension, and loading onto the bead-chips in hybridization chamber inserts. The bead-chips were incubated in a hybridization oven at 48°C for 16-24 hours, washed and prepared for single-base extension, followed by staining with fluorescent dyes and scanning. The data generated was checked for quality through a custom Bioinformatics pipeline, MG-Array QC pipeline and QC passed data was used for CAD KARDIOGEN Risk Score prediction.

PREDICTING KARDIOGEN RISK SCORE USING LDPRED ALGORITHM

Genotype imputation was performed through MedGenome's proprietary pipeline, assessed for Linkage Disequilibrium, LD sites, Minor Allele Frequencies (MAF) and KARDIOGEN Risk Score was generated as described by Khera et al [8]. The statistical model and the imputation workflow have been optimized and validated on South Asian population. The details of the methodology is available in our publication in *Journal of the American College of Cardiology* [19]. Samples were divided into 10 groups (deciles) based on their KARDIOGEN Risk Score, where decile 1 is the lowest KARDIOGEN Risk Score and decile 10 being the highest.

The KARDIOGEN Risk Score cut-off for each decile was based on Control samples decile thresholds. Odds ratio for each decile was calculated with reference to the middle deciles (decile 4 to 6).

Ram Murthy	Ramesh Menon
Dr. Ram Murthy Anjanappa, PhD Senior Scientist	**Dr. Ramesh Menon,** PhD Associate Director

LIMITATIONS

Genetic testing is an important part of the diagnostic process.
However, genetic tests may not always give a definitive answer. This may be due to limitations in current medical knowledge or testing technology. Genetic testing is highly accurate. Rarely, inaccurate results may occur for various reasons. These reasons include, but are not limited to: mislabelled samples, inaccurate reporting of clinical/medical information, rare technical errors or unusual circumstances such as bone marrow transplantation, blood transfusion; or the presence of change(s) in such a small percentage of cells that may not be detectable by the test (mosaicism).

DISCLAIMERS

- CAD- KARDIOGEN Risk Score interpretation is based on a statistical model created from GWAS data from Caucasian populations which was validated on the South Asian population. Ethnicity and environmental factors play a key role in the manifestation of complex diseases like CAD.
- The polygenic risk score must be correlated with the individual's age, family history and lifestyle,
  i.e. smoking, exercise, eating habits, obesity and presence of comorbid medical conditions such as hypertension and diabetes. This is a screening test and not a diagnostic test. It does not confirm the presence or absence of Coronary Artery Disease. Other diagnostic tests may still need to be done to diagnose the presence of the disease. Please consult your cardiologist for further details.
- The report shall be generated within turnaround time (TAT), however, such TAT may vary depending upon the complexity of test(s) requested. MedGenome, under no circumstances, will be liable for any delay beyond the aforementioned TAT.
- It is hereby clarified that the report(s) generated from the test(s) do not provide any diagnosis or opinion or recommends any cure in any manner. MedGenome hereby recommends the patient to take assistance of the clinician or a certified physician or doctor, to interpret the report(s) thus generated. MedGenome hereby disclaims all liability arising in connection with the report(s).
- In a very few cases genetic test may not show the correct results, e.g. because of the quality of the material provided to MedGenome. In case any test provided by MedGenome fails for unforeseeable or unknown reasons, that cannot be influenced by MedGenome in advance, MedGenome shall not be responsible for the incomplete, potentially misleading or even wrong result of any testing if such could not be recognized by MedGenome in advance.

REFERENCES

1. 1000 Genomes Project Consortium et al. A global reference for human genetic variation. *Nature* 2015; 526(7571):68-74.
2. Torkamani A., et al. The personal and clinical utility of polygenic risk scores. *Nat Rev Genet* 2018; 19(9); 581-590.
3. http://www.cardiogramplusc4d.org/publications/
4. http://www.usc.es/cegen/wp-content/uploads/2017/06/infinium-commercial-gsa-data-sheet-370-2016-016.pdf
5. http://www.who.int/macrohealth/action/Report%20of%20the%20National%20Commission.pdf
6. https://www.framinghamheartstudy.org/
7. Khera AV., et al. Genetic risk, adherence to a healthy lifestyle, and coronary disease. *N Engl J Med* 2016; 375 (24): 2349-2358.
8. Khera AV., et al. Genome-Wide Polygenic Scores for Common Diseases Identify Individuals with Risk Equivalent to Monogenic Mutations. *Nat Genet* 2018; 50 (9): 1219-1224.
9. Lvovs D., et al. A polygenic approach to the study of polygenic diseases. *Acta Naturae* 2012; 4 (3):59-71.
10. Nikpay M., et al. A comprehensive 1,000 Genomes-based genome-wide association meta-analysis of coronary artery disease. *Nat Genet* 2015; 47(10):1121-1130.
11. Prabhakaran D., et.al. Global Burden of Cardiovascular Disease, Cardiovascular Diseases in India: Current epidemiology andFuture Directions. *Circulation* 2016; 133 (16):1605-1620.
12. Richards S., et al. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genet Med* 2015; 17(5):405-424.
13. Sherry ST., et al. dbSNP: the NCBI database of genetic variation. *Nucleic Acids Res* 2001; 29(1):308-311.
14. Stenson PD., et al. The Human Gene Mutation Database: towards a com prehensive repository of inherited mutation data for medical research, genetic diagnosis and next-generation sequencing studies. *Human Genet* 2017; 136(6): 665-677.
15. Tada H., et al. Risk prediction by genetic risk scores for coronary heart disease is independent of self-reported family history. *Eur Heart J* 2015; 37(6) 561-567.
16. Thériault S., et al. Polygenic Contribution in Individuals with Early- Onset Coronary Artery Disease Clinical Perspective. *Circ Genom Precis Med* 2018; 11 (1): e001849.
17. Welter D., et al. The NHGRI GWAS Catalog, a curated resource of SNP-trait associations. *Nucleic Acids Res* 2014; 42 (Database issue):D1001-1006.
18. Khera AV., et al. Whole-Genome Sequencing to Characterize Monogenic and Polygenic Contributions in Patients Hospitalized With Early-Onset Myocardial Infarction. *Circulation* 2019; 139(13):1593-1602.
19. Wang M., et al. Validation of a Genome-Wide Polygenic Score for Coronary Artery Disease in South Asians. *J Am Coll Cardiol* 2020; 76(6):703-714.

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